THE GLOBAL SECTIONS OF THE SHEAF OF RELATIVE DIFFERENTIALS OVER AN AFFINE CHART ARE THE MODULE OF DIFFERENTIALS OF THE CORRESPONDING RING MAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Restriction of the sheaf of relative differentials to open subschemes, cited from the same chapter. Cited at proof lines 6669-6670. Lets the proof replace X and S by the affine opens U and V and work with a ring map.
- The universal property of the module of differentials and the affine description of derivations of a module sheaf. Cited at proof lines 6686-6688 and 6691-6692, the second being a card of this lane (L4005). Identifies the module at hand with the module of differentials of the ring map once it is shown to satisfy the universal property.
- Yoneda's lemma and the universal property of the algebra module of differentials, cited from the categories and algebra chapters. Cited at proof lines 6698-6700. Converts the agreement of the represented functors into the asserted unique isomorphism.

Source reference. Original lemma 00RO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1010–1010. Same-chapter reference used in the retained author text at source line(s) 6481. It is supplied only as a navigation pointer

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Source reference. Original lemma 001P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 70–70. Same-chapter reference used in the retained author text at source line(s) 6482. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2417–2417. Same-chapter reference used in the retained author text at source line(s) 6465. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01US: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2421–2421. Same-chapter reference used in the retained author text at source line(s) 6452. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2420–2420. Same-chapter reference used in the retained author text at source line(s) 6464. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original section 01I6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1967–1967. Same-chapter reference used in the retained author text at source line(s) 6459. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. For any pair of affine opens $\mathrm{Spec}(A) = U \subset X$, $\mathrm{Spec}(R) = V \subset S$ with $f(U) \subset V$ there is a unique isomorphism*

$$\Gamma(U, \Omega_{X/S}) = \Omega_{A/R}.$$

compatible with $d_{X/S}$ and $d : A \to \Omega_{A/R}$.

Proof. By Lemma 01US we may replace X and S by U and V. Thus we may assume $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(R)$ and we have to show the lemma with $U = X$ and $V = S$. Consider the A-module $M = \Gamma(X, \Omega_{X/S})$ together with the R-derivation $\mathrm{d}_{X/S} : A \to M$. Let N be another A-module and denote $\widetilde{N}$ the quasi-coherent $\mathcal{O}_X$-module associated to N, see Schemes, Section 01I6. Precomposing by $\mathrm{d}_{X/S} : A \to M$ we get an arrow

$$\alpha : \mathrm{Hom}_A(M, N) \longrightarrow \mathrm{Der}_R(A, N)$$

Using Lemmas 01UR and 01UO we get identifications

$$\mathrm{Hom}_{\mathcal{O}_X}(\Omega_{X/S}, \widetilde{N}) = \mathrm{Der}_S(\mathcal{O}_X, \widetilde{N}) = \mathrm{Der}_R(A, N)$$

Taking global sections determines an arrow $\mathrm{Hom}_{\mathcal{O}_X}(\Omega_{X/S}, \widetilde{N}) \to \mathrm{Hom}_R(M, N)$. Combining this arrow and the identifications above we get an arrow

$$\beta : \mathrm{Der}_R(A, N) \longrightarrow \mathrm{Hom}_R(M, N)$$

Checking what happens on global sections, we find that α and β are each others inverse. Hence we see that $\mathrm{d}_{X/S} : A \to M$ satisfies the same universal property as $\mathrm{d} : A \to \Omega_{A/R}$, see Algebra, Lemma 00RO. Thus the Yoneda lemma (Categories, Lemma 001P) implies there is a unique isomorphism of A-modules $M \cong \Omega_{A/R}$ compatible with derivations. □